# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Scenario-Based Research Assessment

**COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01**

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**COURSE OUTCOME COVERED**

**CO1:** *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

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**Total Marks: 50** **Time: 60 Mins**

**No. of Questions: 5**

Annexure A

## Scenario-Based Research Assessment

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***Code of Conduct:***

*I Maniyam Anushka(192525411) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:Maniyam Anushka***

| Criterion | Weightage | Marks Obtained |
|---|---|---|
| **Identification of Latest AI Application** | 10% | |
| **Problem Analysis and Domain Understanding** | 15% | |
| **AI Technologies and Working Mechanism** | 20% | |
| **Research Quality and Literature Review** | 10% | |
| **Real-World Implementation and Case Analysis** | 10% | |
| **Impact, Challenges, and Future Scope** | 15% | |
| **Originality and Critical Thinking** | 10% | |
| **Organization, Presentation, and Documentation** | 10% | |
| **Total** | 100% | |

Signature of the Faculty Total Marks Awarded

## ANSWER ANY ONE

| No. | AI Domain | Scenario-Based Research Question |
|---|---|---|
| 1 | **Healthcare** | Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations. |
| 2 | **Finance** | Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges. |
| 3 | **Retail** | Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope. |
| 4 | **Autonomous Vehicles** | The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges. |
| 5 | **Robotics** | Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments. |
| 6 | **Manufacturing** | Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations. |

**Structure of the Report:**

| S.No | Report Component | Student Deliverable / Guiding Question |
|---|---|---|
| 1 | **Title of AI Application** | Provide the name of the latest AI application selected and the corresponding AI domain. |
| 2 | **Domain Selection** | Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing). |
| 3 | **Introduction and Background** | Describe the AI application and provide an overview of its purpose and importance. |
| 4 | **Problem Identification** | Explain the real-world problem or challenge addressed by this AI application. |
| 5 | **Need for AI Solution** | Analyze why AI is required and how it improves traditional approaches. |
| 6 | **AI Technologies Used** | Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.). |
| 7 | **Working Mechanism** | Explain how the AI system works, including input data, processing methods, decision-making, and output generation. |
| 8 | **Data Sources and Processing** | Describe the type of data used and how the data is processed for intelligent decision-making. |
| 9 | **Real-World Implementation** | Identify organizations, industries, or real-world scenarios where this AI application is currently implemented. |
| 10 | **Impact and Benefits** | Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation. |
| 11 | **Challenges and Limitations** | Discuss technical, ethical, privacy, security, and deployment challenges associated with the application. |
| 12 | **Future Scope and Emerging Trends** | Explain possible enhancements, future developments, and research opportunities in this AI application. |
| 13 | **Conclusion** | Summarize the significance of the AI application and its contribution to solving real-world problems. |
| 14 | **References** | Provide research papers, technical articles, industry reports, and reliable sources referred for the study. |

### Rubrics for Evaluation

| S.No | Criteria | Weightage (%) | Excellent (100–90%) | Good (89–75%) | Satisfactory (74–60%) | Improvement ( |
|---|---|---|---|---|---|---|
| 1. | Identification of Latest AI Application | 10% | Selects a highly relevant, emerging AI application with strong justification. | Selects a recent and relevant AI application. | Selects a suitable application with limited justification. | Application is outdated or irrelevant. |
| 2. | Problem Analysis and Domain Understanding | 15% | Clearly explains the real-world problem, domain context, and need for AI with critical analysis. | Good explanation with adequate analysis. | Basic explanation with limited analysis. | Problem statement is unclear or incomplete. |
| 3. | AI Technologies and Working Mechanism | 20% | Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth. | Good explanation with minor omissions. | Covers only basic AI concepts. | AI technologies are poorly explained or incorrect. |
| 4. | Research Quality and Literature Review | 10% | Uses recent, credible sources with proper citations and excellent synthesis. | Good quality references with appropriate citations. | Limited references with basic discussion. | Few or unreliable references; poor citation. |
| 5. | Real-World Implementation and Case Analysis | 10% | Thoroughly analyzes industrial implementation with appropriate examples. | Good implementation analysis. | Basic implementation discussion. | Minimal or no real-world analysis. |
| 6. | Impact, Challenges, and Future Scope | 15% | Critically evaluates benefits, limitations, ethical issues, and future trends. | Good evaluation with minor gaps. | Basic discussion of impact and future scope. | Limited evaluation or missing sections. |
| 7. | Originality and Critical Thinking | 10% | Demonstrates independent research, innovation, and critical evaluation. | Good analytical thinking. | Some originality with limited analysis. | Mostly descriptive or copied content. |
| 8. | Organization, Presentation, and Documentation | 10% | Exceptionally organized, professional presentation, error-free, and properly referenced. | Well organized with minor issues. | Adequately organized with some errors. | Poor organization and documentation. |

## 1. Title of AI Application

**AI-Assisted Radiology Diagnosis Systems** — Artificial Intelligence tools that read X-rays, CT scans, MRI scans and mammograms to detect diseases such as cancer, stroke, fractures and lung conditions, and to flag urgent cases for doctors.

**AI Domain:** Healthcare (Medical Imaging / Radiology AI)

## 2. Domain Selection

The chosen domain is **Healthcare**, and specifically the sub-area of **medical imaging and radiology**. This area was selected because radiology is the single largest area of AI adoption in medicine today, and new AI tools are being cleared for hospital use almost every month, making it an excellent example of a fast-growing, real-world AI application.

## 3. Introduction and Background

Radiology involves examining medical images to identify signs of disease. Traditionally, this is done entirely by human radiologists, who study each scan carefully and write a report. As hospitals produce thousands of scans every day, this process can be slow and tiring, and small abnormalities are sometimes missed.

AI-assisted radiology systems use deep learning models, especially Convolutional Neural Networks (CNNs), to automatically analyse medical images. These systems can highlight suspicious areas, measure abnormalities, and even rank patients by urgency so that the most critical cases are reviewed first. Well-known examples include Aidoc (body CT triage), Viz.ai (stroke detection), Qure.ai (chest X-ray screening), and BoneView (fracture detection).

This field has grown extremely quickly. By early 2026, radiology accounted for about three-quarters of all AI-based medical devices cleared by the U.S. Food and Drug Administration (FDA), making it by far the leading specialty for AI use in medicine.

## 4. Problem Identification

The real-world problems addressed by this AI application are:

- A global shortage of trained radiologists, especially in rural areas and developing countries.
- A very large and increasing volume of scans, which causes reporting delays.
- Human fatigue and the risk of missing small but serious findings, such as an early tumour or a hairline fracture.
- Time-critical emergencies, such as stroke, where every minute of delay can cause permanent harm to the patient.
- Inconsistent second opinions between different radiologists reading the same scan.

## 5. Need for AI Solution

Traditional (manual) reading of scans depends completely on the availability and alertness of a human expert. AI is needed because it can:

- Scan images in seconds, much faster than a human, and work 24 hours a day without fatigue.
- Act as a tireless "second reader" that checks every image for findings a doctor might otherwise miss.

- Automatically re-order the worklist so that the most urgent case (for example, a stroke or internal bleeding) is opened first, instead of waiting in a normal queue.
- Bring specialist-level screening to clinics that do not have enough radiologists, particularly in low-resource regions.

In short, AI does not aim to replace the radiologist, but to support faster and more consistent decisions, especially under time pressure or staff shortage.

## 6. AI Technologies Used

| AI Technique | How it is Used in Radiology |
|---|---|
| Deep Learning (CNNs) | Core technology for recognising patterns in X-ray, CT and MRI images, su h as tumours, |
| Computer Vision | Detects, segments and measures abnormal regions inside an image (e.g. outlining a lung |
| Vision-Language Models | Newer models (such as NVIDIA Clara Reason) that not only detect a findi g but also gen |
| Machine Learning (Risk Scoring) | Used in models such as MIT's Mirai to predict a patient's future risk of dise se (e.g. brea |
| Natural Language Processing (NL P H) | elps in drafting or summarising the final radiology report from the AI's fin ings. |

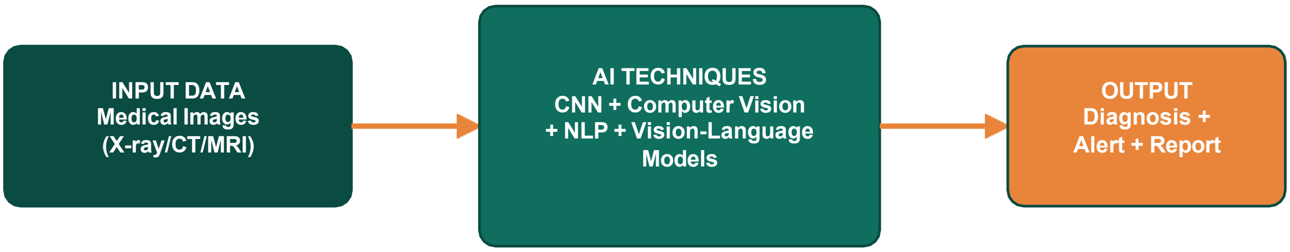


*Figure 1: End-to-end AI pipeline — from medical image input to diagnostic output.*

## 7. Working Mechanism

An AI radiology system generally works in the following steps:

- **Input:** A scan (X-ray, CT or MRI) is automatically sent from the hospital's imaging system (PACS) to the AI software as soon as it is captured.
- **Pre-processing:** The image is resized, normalised and cleaned so the AI model can read it correctly.
- **Feature extraction:** A trained CNN scans the image pixel by pixel to find patterns that match known signs of disease, using layers of filters learned from millions of prior images.
- **Decision-making:** The model outputs a probability score (for example, "92% chance of large vessel occlusion stroke") and marks the exact location of the finding on the image.
- **Output and alert:** If the finding is urgent, the system instantly sends a smartphone or PACS alert to the radiologist and care team; otherwise, it adds the case to the normal worklist along with the AI's marked image and confidence score.
- **Human review:** The radiologist always reviews the AI's suggestion, confirms or corrects it, and writes the final medical report.

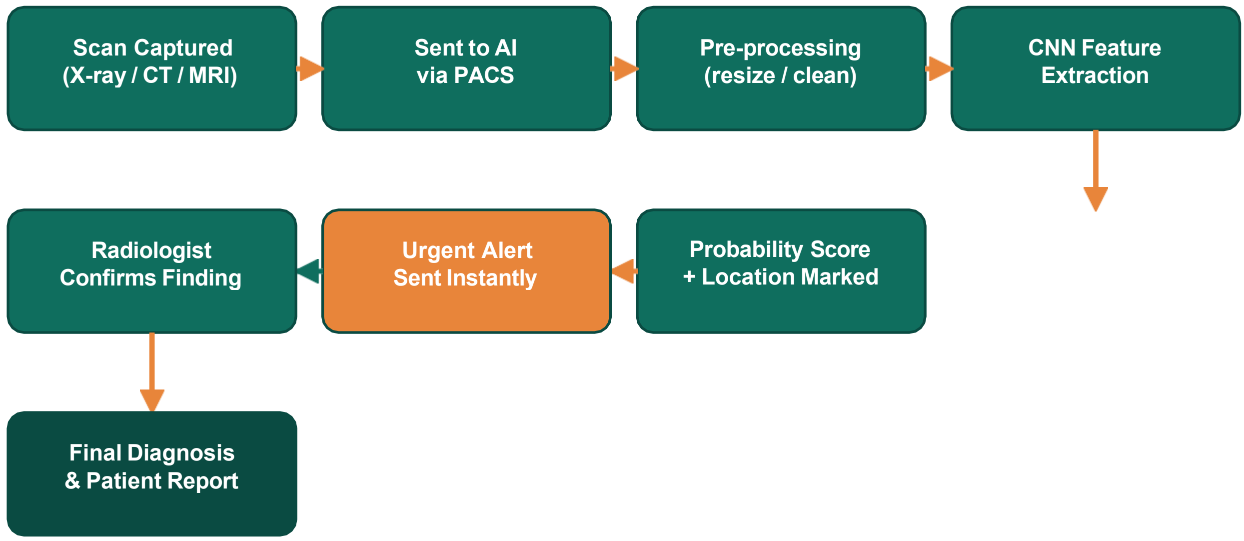


*Figure 2: Flowchart of the AI radiology working mechanism, from scan capture to final diagnosis.*

## 8. Data Sources and Processing

These systems are trained using very large sets of labelled medical images, such as millions of chest X-rays, CT scans and MRI scans, each marked by expert radiologists to show what disease (if any) is present. For example, one widely used chest X-ray model was trained on more than nine million exams.

During training, the data is processed in three broad stages: (1) cleaning and labelling the images, (2) training the deep learning model to recognise patterns linked to each label, and (3) testing the model on new, unseen images to check its accuracy before it is approved for hospital use. Patient data is anonymised, and strict privacy rules (such as HIPAA in the United States) apply throughout.

## 9. Real-World Implementation

AI radiology tools are now used in thousands of hospitals worldwide. Some notable examples:

- **Viz.ai** detects stroke-causing blood-vessel blockages from CT scans and instantly alerts stroke teams; it is deployed in more than 1,700 hospitals.
- **Aidoc** received FDA clearance in 2026 for a single AI tool that screens body CT scans for 14 different urgent conditions in one pass, such as appendicitis and internal bleeding.
- **Qure.ai's qXR** is used for high-volume chest X-ray screening, including at University Hospitals in Ohio, and is especially valuable in countries with very few radiologists.
- **BoneView** assists emergency doctors in detecting fractures on X-rays and has received both European (CE) and FDA clearance.
- By March 2026, the FDA had cleared over 1,500 AI medical devices in total, and radiology made up roughly three-quarters of them — showing how widely this technology has already been adopted.

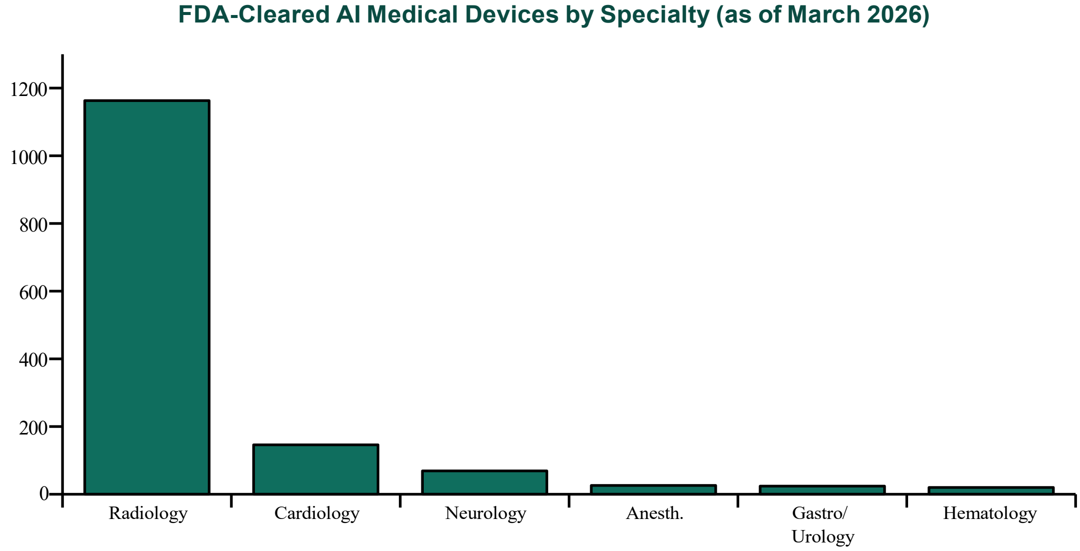


*Figure 3: Radiology leads every other medical specialty by a wide margin in FDA-cleared AI devices.*

## 10. Impact and Benefits

- **Speed:** Some chest X-ray AI tools cut reporting turnaround time by up to 40%.
- **Accuracy:** Recent FDA-cleared tools report very high performance, for example around 97% sensitivity and 98% specificity for detecting certain urgent body-CT conditions.
- **Better triage:** Life-threatening cases such as stroke are flagged and pushed to the front of the queue, helping doctors act within the critical time window.
- **Wider access:** AI screening brings expert-level image checking to clinics that lack enough radiologists, improving healthcare access in underserved areas.
- **Support, not replacement:** Radiologists are freed from repetitive screening tasks and can spend more time on complex cases and patient-facing procedures.

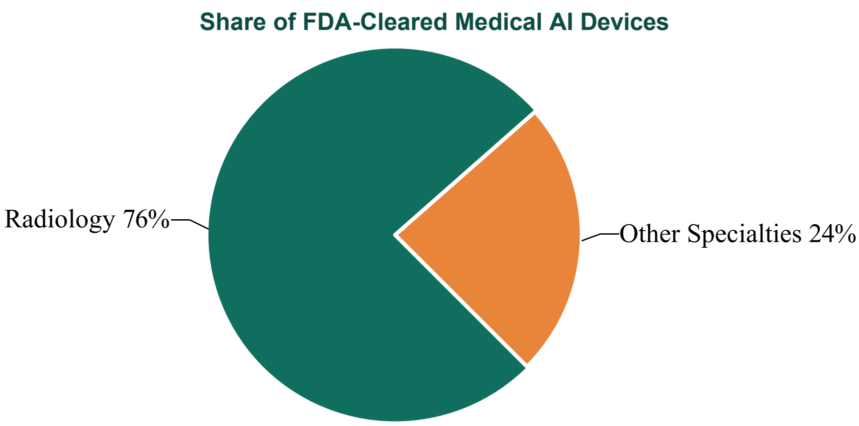


*Figure 4: Radiology's dominant share of all FDA-cleared medical AI devices highlights how central imaging AI has become to healthcare.*

## 11. Challenges and Limitations

- **Low real-world adoption:** Even though over a thousand tools are approved, studies suggest only around 30% of radiologists actually use AI in daily practice, showing a gap between approval and real use.
- **Bias and generalisability:** A model trained mostly on one type of scanner, hospital, or population may perform poorly on different equipment or patient groups.
- **Explainability:** Many deep learning models act like a "black box", making it hard for doctors to understand exactly why the AI flagged a case — a newer generation of models is trying to fix this by giving step-by-step reasoning.
- **Regulatory and legal responsibility:** It is not always clear who is responsible for a missed diagnosis — the hospital, the AI vendor, or the doctor.
- **Cost and reimbursement:** Most hospitals cannot yet claim insurance payment for AI-aided reads, apart from a few areas like cardiac imaging, which discourages wider purchase of these tools.
- **Data privacy and security:** Handling sensitive patient scans requires strong cybersecurity and strict compliance with health-data protection laws.

## 12. Future Scope and Emerging Trends

- **Foundation and vision-language models:** Newer AI models can both detect a finding and explain it in plain text, moving radiology AI closer to an assistant that "talks" like a doctor.
- **Multi-condition screening:** Instead of one tool per disease, single AI systems are starting to screen for many conditions from one scan.
- **Stronger regulation:** Frameworks such as the EU AI Act (expected to fully apply to high-risk radiology AI by 2026) will require clearer documentation of training data, bias checks, and human oversight.
- **Better reimbursement pathways:** More insurance billing codes are expected to be created for AI-assisted reads, which should encourage wider hospital adoption.
- **Personalised risk prediction:** Models like MIT's Mirai point toward AI that predicts a patient's future disease risk, not just current findings, enabling earlier prevention.

## 13. Conclusion

AI-assisted radiology diagnosis is one of the most mature and widely deployed applications of Artificial Intelligence in healthcare today. By using deep learning to read medical images quickly and consistently, these tools help doctors catch time-critical conditions such as stroke earlier, reduce reporting delays, and extend expert-level screening to places with few radiologists. At the same time, real challenges remain around trust, bias, explainability, and payment models. As regulation matures and newer explainable AI models emerge, AI is expected to become a standard "co-pilot" for radiologists rather than a replacement — ultimately making diagnosis faster, more accurate, and more widely available.

## 14. References

1 IntuitionLabs. "AI in Radiology: 2025 Trends, FDA Approvals & Adoption." intuitionlabs.ai, 2025.

2 Radiology Business. "Radiology gets 68 new FDA-cleared algorithms." radiologybusiness.com, 2026.

3 The Imaging Wire. "FDA Updates AI List with New Clearances." theimagingwire.com, 2026.

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5 BuildMVPFast. "FDA-Cleared AI Radiology Tools Hospitals Actually Use." buildmvpfast.com, 2026.

6 Pinggy. "AI Medical Imaging in 2026: Best Radiology AI Tools, FDA Clearances, and Diagnostic Accuracy." pinggy.io, 2026.

7 National Center for Biotechnology Information (NCBI/PMC). "Artificial Intelligence in Fracture Diagnosis on Radiographs: Evidence, Pitfalls, and Pathways for Clinical Integration (2020-2025)." ncbi.nlm.nih.gov, 2025.